

Healthcare Admissions SQL Case Study

Introduction

Hi, my name is Gia. In this case study, I analyze a healthcare admissions dataset using SQL to examine billing patterns, admission types, length of stay, and insurance coverage. The objective of this analysis is to identify key cost drivers and utilization trends within hospital admissions while demonstrating practical SQL analysis skills in a healthcare context.

Business Objective

The objective of this analysis is to evaluate healthcare admissions and hospital utilization by addressing the following questions:

- How much total billing is generated across all hospital admissions?
- Which medical conditions contribute most to overall healthcare costs?
- How do billing patterns differ by admission type?
- Does length of stay vary by admission type or medical condition?
- Are there meaningful differences in billing patterns across insurance providers?

Data Source

The dataset used in this analysis comes from a **synthetic healthcare admissions dataset** available on Kaggle. Each row represents a single hospital admission and includes patient, billing, and admission-related information.

Data Description

Key fields used:

- Medical Condition
- Admission Type
- Hospital
- Insurance Provider
- Billing Amount
- Date of Admission
- Discharge Date

Data Credibility (ROCCC-style)

Reliable: Publicly available synthetic healthcare dataset used for analysis practice

Original: Provided by the dataset author on Kaggle

Comprehensive: Includes patient-level admission, billing, and length-of-stay data

Current: Represents recent healthcare admission patterns

Cited: Dataset source is clearly documented

PROCESS / CLEAN

Before analysis, the dataset was reviewed to ensure basic data quality and consistency:

- Confirmed total admission record count
- Verified numeric fields (billing amount, age, room number)
- Ensured admission and discharge dates were usable for time-based analysis
- Standardized table naming for clarity during analysis

ANALYZE

Dataset Summary:

An initial SQL query was used to confirm the size and structure of the healthcare admissions dataset.

Key finding:

The dataset contains 55,500 hospital admission records, spanning 39,876 distinct hospitals, 5 insurance providers, and 6 medical conditions.

```
ALTER TABLE healthcare_dataset RENAME TO healthcare;
```

```
SELECT COUNT(*) AS total_rows
FROM healthcare;
```

123 total_rows
55,500

```
SELECT
COUNT(DISTINCT Hospital) AS hospitals,
COUNT(DISTINCT "Insurance Provider") AS insurance_providers,
COUNT(DISTINCT "Medical Condition") AS medical_conditions
FROM healthcare;
```

123 hospitals	123 insurance_providers	123 medical_conditions
39,876	5	6

Overall Billing Summary:

Overall billing was analyzed to understand the scale and range of healthcare costs across hospital admissions.

Findings:

- Total billing is approximately \$1.42B.
- Average billing per admission is about \$25.5K.
- Billing amounts vary widely across admissions.
- A small number of negative billing values appear, likely due to adjustments or synthetic data artifacts.

```
SELECT
  COUNT(*) AS total_admissions,
  ROUND(SUM("Billing Amount"), 2) AS total_billing,
  ROUND(AVG("Billing Amount"), 2) AS avg_billing,
  ROUND(MIN("Billing Amount"), 2) AS min_billing,
  ROUND(MAX("Billing Amount"), 2) AS max_billing
FROM healthcare;
```

123 total_admissions ▼	123 total_billing ▼	123 avg_billing ▼	123 min_billing ▼
55,500	1,417,432,043.4	25,539.32	-2,008.49

Cost drivers by medical condition:

This analysis compares billing patterns across medical conditions to identify key cost drivers.

Findings:

- Diabetes has the highest total billing ~\$238.5M.
- Obesity and arthritis follow closely.
- Average billing per patient is similar across conditions ~\$25K–\$25.8K.
- Total cost differences are driven by patient volume.

```
SELECT
  "Medical Condition" AS medical_condition,
  COUNT(*) AS patients,
  ROUND(SUM("Billing Amount"), 2) AS total_billing,
  ROUND(AVG("Billing Amount"), 2) AS avg_billing
FROM healthcare
GROUP BY "Medical Condition"
ORDER BY total_billing DESC;
```

	A-Z medical_condition	123 patients	123 total_billing	123 avg_billing
1	Diabetes	9,304	238,539,725.49	25,638.41
2	Obesity	9,231	238,214,920.69	25,805.97
3	Arthritis	9,308	237,329,120.23	25,497.33
4	Hypertension	9,245	235,720,650.31	25,497.1
5	Asthma	9,185	235,459,765.36	25,635.25
6	Cancer	9,227	232,167,861.31	25,161.79

Admission type analysis:

Billing was grouped by admission type to compare cost patterns across elective, urgent, and emergency visits.

Findings:

- Elective admissions generate the highest total billing ~\$477.6M.
- Urgent and emergency admissions follow closely in total cost.
- Average billing per admission is similar across all admission types ~\$25.5K.
- Differences in total billing are driven primarily by patient volume.

```
SELECT
  "Admission Type" AS admission_type,
  COUNT(*) AS patients,
  ROUND(SUM("Billing Amount"), 2) AS total_billing,
  ROUND(AVG("Billing Amount"), 2) AS avg_billing
FROM healthcare
GROUP BY "Admission Type"
ORDER BY total_billing DESC;
```

A-Z admission_type	123 patients	123 total_billing	123 avg_billing
Elective	18,655	477,609,531.84	25,602.23
Urgent	18,576	474,010,562.9	25,517.36
Emergency	18,269	465,811,948.66	25,497.4

Length of stay:

Average length of stay was analyzed to evaluate hospital utilization across different admission types.

Findings:

- Emergency admissions have the longest average stay (~15.6 days).
- Elective admissions average ~15.5 days.

- Urgent admissions average ~15.4 days.
- Differences in length of stay across admission types are minimal.

```
SELECT
  "Admission Type" AS admission_type,
  ROUND(
    AVG(
      julianday("Discharge Date") - julianday("Date of Admission")
    ), 2
  ) AS avg_length_of_stay_days
FROM healthcare
GROUP BY "Admission Type";
```

A-Z admission_type ▼	123 avg_length_of_stay_days ▼
Elective	15.53
Emergency	15.6
Urgent	15.41

Length of stay by medical condition (min 1000 patients):

Length of stay was analyzed by medical condition to compare inpatient utilization patterns.

Findings:

- Asthma patients have the longest average stay (~15.7 days).
- Arthritis and cancer follow closely.
- Diabetes patients have the shortest average stay (~15.4 days).
- Overall differences in length of stay across conditions are small.

```
SELECT
  "Medical Condition" AS medical_condition,
  COUNT(*) AS patients,
  ROUND(
    AVG(julianday("Discharge Date") - julianday("Date of Admission")),
    2
  ) AS avg_length_of_stay_days
FROM healthcare
GROUP BY "Medical Condition"
HAVING COUNT(*) >= 1000
ORDER BY avg_length_of_stay_days DESC;
```

A-Z medical_condition ▼	123 patients ▼	123 avg_length_of_stay_days ▼
Asthma	9,185	15.7
Arthritis	9,308	15.52
Cancer	9,227	15.5
Obesity	9,231	15.46
Hypertension	9,245	15.46
Diabetes	9,304	15.42

Insurance provider distribution:

Billing was analyzed by insurance provider to evaluate differences in patient volume and cost patterns.

Findings:

- Patient counts are evenly distributed across the five insurance providers.
- Total billing amounts are similar across providers.
- Average billing per patient shows minimal variation ~\$25K–\$25.6K.
- Differences in total billing are driven by volume rather than payer type.

```
SELECT
  "Insurance Provider" AS insurance_provider,
  COUNT(*) AS patients,
  ROUND(SUM("Billing Amount"), 2) AS total_billing,
  ROUND(AVG("Billing Amount"), 2) AS avg_billing
FROM healthcare
GROUP BY "Insurance Provider"
ORDER BY patients DESC; |
```

A-Z insurance_provider ▼	123 patients ▼	123 total_billing ▼	123 avg_billing ▼
Cigna	11,249	287,139,345.27	25,525.77
Medicare	11,154	285,720,758.13	25,615.99
UnitedHealthcare	11,125	282,454,542.84	25,389.17
Blue Cross	11,059	283,254,294.21	25,613.01
Aetna	10,913	278,863,102.95	25,553.29

CONCLUSIONS

This analysis highlights how healthcare costs and utilization patterns are influenced more by patient volume than by large differences in per-patient billing or length of stay. Across medical conditions, admission types, and insurance providers, billing and utilization metrics remain relatively consistent, suggesting standardized care and pricing structures within the dataset.

Tools Used

SQLite

DBeaver

GitHub (project documentation)